1. (a)	$p^{\frac{1}{2}} = \left(\frac{1}{16}x^4\right)^{\frac{1}{2}} = \frac{1}{4}x^2$	B1 **(1)**
(b)	$(pq)^{-1} = \left(\frac{1}{16}x^4 \times \frac{40}{x^3}\right)^{-1} = \left(\frac{5}{2}x\right)^{-1} = \frac{2}{5}x^{-1}$	M1, A1 **(2)**
(c)	$p\,q^2 = \frac{1}{16}x^4 \times \left(\frac{40}{x^3}\right)^2 = \frac{1600}{16} \times \frac{x^4}{x^6} = 100x^{-2}$	M1, A1 **(2)**
		(5 marks)

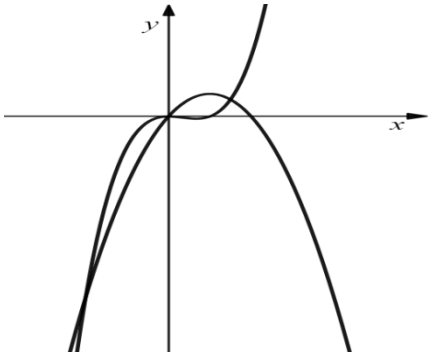
	$y = 2x^{\frac{5}{2}} - 4x + 3$	
2. (a)	$\left\{\frac{dy}{dx} =\right\} 5x^{\frac{3}{2}} - 4$	M1, A1 **(2)**
(b)	$5x^{\frac{3}{2}} - 4 = 16 \Rightarrow x^{\frac{3}{2}} = 4$ $\Rightarrow x = 4^{\frac{2}{3}} = 2^{\frac{4}{3}} \qquad \Rightarrow k = \frac{4}{3}$	M1, A1 A1 **(3)** **(5 marks)**

3.	$\int \frac{(x+3)^2}{3\sqrt{x}}\,\mathrm{d}x$ $\frac{(x+3)^2}{3\sqrt{x}} = \frac{x^2+6x+9}{3\sqrt{x}} = \frac{1}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$ $\int \frac{(x+3)^2}{3\sqrt{x}}\,\mathrm{d}x = \frac{1}{3}\times\frac{2}{5}x^{\frac{5}{2}} + 2\times\frac{2}{3}x^{\frac{3}{2}} + 3\times\frac{2}{1}x^{\frac{1}{2}} + c$ $= \frac{2}{15}x^{\frac{5}{2}} + \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + c$	M1 dM1 A1, A1, A1 **(5)** **(5 marks)**

4. (a)	$(270°, 0)$	B1 **(1)**
(b)	$(-180°, -4)$	B1, B1 **(2)**
(c)(i)	100	B1
(ii)	4	B1
(iii)	7	B1 **(3)** **(6 marks)**

	$x-2y+25=0$	
5. (a)	$y=\frac{1}{2}x+\frac{25}{2}\Rightarrow y=-2x+...$ $y=-2x$	M1 A1 **(2)**
(b)	Substitutes $y=-2x$ into $x-2y+25=0\Rightarrow x+4x+25=0$ $\Rightarrow x=-5,\ y=10$	M1 A1, A1 **(3)**
(c)	Shortest distance = $\sqrt{5^2+10^2}=5\sqrt{5}$	M1, A1cso **(2)**
		(7 marks)

6 (a)	$\frac{\sin CAO}{17} = \frac{\sin 0.6}{15} \Rightarrow CAO = 0.6944...$	M1
	Angle $COA = \pi - 0.6 - "0.6944" = 1.847$ *	dM1, A1* **(3)**
(b)	Attempts $\frac{1}{2}r^2\theta = \frac{1}{2}\times 15^2 \times \theta$ where $\theta = \left(2\pi - 1.847\right)$ or just 1.847 OR attempts $\frac{1}{2}ab\sin C = \frac{1}{2}\times 15\times 17\sin\left(1.847\right)$	M1
	Attempts $\frac{1}{2}r^2\theta = \frac{1}{2}\times 15^2 \times \left(2\pi - 1.847\right)\left(\approx 499\right)$ AND $\frac{1}{2}ab\sin C = \frac{1}{2}\times 15\times 17\sin\left(1.847\right)\left(\approx 122.7\right)$ AND adds	dM1
	(awrt) 622 m^2	A1 **(3)**
(c)	$r\theta = 15\times\theta$ where $\theta = \left(2\pi - 1.847\right)$ or just 1.847 OR $\left(AC^2 =\right) 15^2 + 17^2 - 2\times 15\times 17\cos\left(1.847\right)\left(\approx 653\right)$	M1
	Attempts $r\theta = 15\times\left(2\pi - 1.847\right)\left(= 66.54\right)$ AND $\left\{AC =\right\}\sqrt{15^2 + 17^2 - 2\times 15\times 17\cos\left(1.847\right)}$	dM1
	92.1 + 2 = (awrt) 94.1 m	A1 **(3)** **(9 marks)**

7 (a)	States or implies that $f(x) = \lambda x^2(x-4)$	M1
	Attempts to find λ. E.g. $120 = \lambda 10^2(10-4) \Rightarrow \lambda = ...$	dM1
	$\{f(x) =\} 0.2x^2(x-4)$	A1
		(3)
(b)		
	"Upside down" parabola	M1
	Passing through O and +ve x axis > 4	A1
		(2)
(c)	Sets $1.2x(8-x) =$ their $0.2x^2(x-4)$	B1ft
	$1.2x(8-x) = 0.2x^2(x-4) \Rightarrow x\left(x^2+2x-48\right) = 0$ or $\Rightarrow x^3+2x^2-48x=0$	M1
	$\Rightarrow x = ...,...(,0)$	dM1
	For $x = -8, 0, 6$ OR one of $(-8,-153.6),(6,14.4)$	A1
	All of $(-8,-153.6),(6,14.4),(0,0)$ $\quad \left\{\left(-8,-\frac{768}{5}\right),\left(6,\frac{72}{5}\right),(0,0)\right\}$	A1
		(5)
		(10 marks)

8 (a) (i)	$x=4, \mathrm{f}'(x)=10,\ \mathrm{f}'(x)=3\sqrt{x}+kx^2 \Rightarrow 10=3\sqrt{4}+4^2k \Rightarrow k=...$	M1
	$10=3\times 2+k\times 16 \Rightarrow k=\frac{1}{4}$ *	A1*
(ii)	$x=4, y=12$ on $y=10x+c \Rightarrow 12=10\times 4+c$	M1
	$\Rightarrow c=-28$	A1
		(4)
(b)	$\mathrm{f}''(x)=\frac{3}{2}x^{-\frac{1}{2}}+\frac{1}{2}x$	M1 A1ft
	$\{\Rightarrow \mathrm{f}''(4)\}=\frac{11}{4}$	A1
		(3)
(c)	$\mathrm{f}(x)=2x^{\frac{3}{2}}+\frac{1}{12}x^3+d$	M1, A1ft
	Uses $P(4,12) \Rightarrow 12=2\times 8+\frac{1}{12}\times 4^3+d \Rightarrow d=...$	dM1
	$\{\mathrm{f}(x)=\}\ 2x^{\frac{3}{2}}+\frac{1}{12}x^3-\frac{28}{3}$	A1
		(4)
		(11 marks)

9 (a) (i)	Stretch parallel to the x-axis $\times\frac{1}{2}$ or stretch parallel to the y-axis $\times\sqrt{2}$	M1, A1
(ii)	Translate by the vector $\begin{pmatrix}0\\12\end{pmatrix}$ (or translate up by 12 (units))	M1, A1
		(4)
(b) (i)	$12-\sqrt{x}=\sqrt{2}\sqrt{x}$	
	$12=\left(\sqrt{2}+1\right)\sqrt{x}$	M1
	$\Rightarrow\sqrt{x}=\frac{12}{\sqrt{2}+1}\times\frac{\sqrt{2}-1}{\sqrt{2}-1}=12\left(\sqrt{2}-1\right)$ *	dM1, A1 *
Alt (i)	$12-\sqrt{x}=\sqrt{2x}\Rightarrow\left(12-\sqrt{x}\right)^2=2x\Rightarrow x+24\sqrt{x}-144=0$	M1
	$\Rightarrow\left(\sqrt{x}\right)=\frac{-24\pm\sqrt{24^2-4\times-144}}{2}=-12\pm\frac{12}{2}\sqrt{4+8}=-12\pm12\sqrt{2}$	dM1
	$\sqrt{x}>0\Rightarrow\sqrt{x}=-12+12\sqrt{2}=12\left(\sqrt{2}-1\right)$*	A1
(ii)	$\Rightarrow x=12^2\left(\sqrt{2}-1\right)^2=144\left(2+1-2\sqrt{2}\right)=144\left(3-2\sqrt{2}\right)$	M1, A1
	$y\left\{=12-\sqrt{x}=12-12\left(\sqrt{2}-1\right)\right\}=12\left(2-\sqrt{2}\right)$	B1
	Or common acceptable alt forms: $P\left(432-288\sqrt{2},\quad 24-12\sqrt{2}\right)$	
		(6) **(10 marks)**

10	$(k-1)x^6 + 4x^3 + (k-4) = 0$	
(a)	$3.5x^6 + 4x^3 + 0.5 = 0 \Rightarrow 7x^6 + 8x^3 + 1 = 0$	
	$\Rightarrow \left(x^3 + 1\right)\left(7x^3 + 1\right) = 0$	M1
	$\Rightarrow x^3 = -1, \quad x^3 = -\frac{1}{7}$	A1
	$\Rightarrow x = -1, \quad x = -\frac{1}{\sqrt[3]{7}}$	A1
		(3)
(b)	Attempts $b^2 - 4ac = 16 - 4(k-1)(k-4)$	M1
	$= 20k - 4k^2$	A1
	Solves $b^2 - 4ac < 0 \Rightarrow 4k(5-k) < 0 \Rightarrow k < 0, \ k > 5$	dM1 A1
		(4)
		(7 marks)